

CONQUEST complete test-suite report

Branch feature/general-nonorthogonal-cells
Commit 9248eb4de5f5
Run completed 2026-08-09T23:57:16+09:00

32 passed	0 failed	0 not run
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MPI policy: every launch was capped at $\max(1, \min(3, \text{floor}(\text{number of atoms} / 3)))$. OpenMP was fixed at one thread and the build used no more than three jobs.

Results overview

Test	Atoms	MPI ranks used	Time	Status
test_001_bulk_Si_1proc_Diag	8	2	2.6 s	PASS
test_002_bulk_Si_1proc_OrderN	8	2	9.9 s	PASS
test_003_bulk_BTO_polarisation	5	1	18.1 s	PASS
test_004_isol_C2H4_4proc_PBE0CRI	6	2	12.2 s	PASS
test_005_isol_C2H4_4proc_PBE0GTO	6	2	56.4 s	PASS
test_006_isol_C2H4_4proc_PBE0ERI	6	2	16.4 s	PASS
test_007_isol_CH_spinpol_1proc_PBE0CRI	2	1	228.7 s	PASS
test_008_surface_dipole	3	1	4.5 s	PASS
test_009_DFT_plus_U	2	1	188.4 s	PASS
test_010_graphite_monoclinic	4	1, 2	194.1 s	PASS
test_011_hfo2	12	1, 2	63.2 s	PASS
test_012_bulk_Si_primitive_nonorthogonal	2	1	47.4 s	PASS
test_013_rutile_TiO2_tetragonal	6	1, 2	153.7 s	PASS
test_014_bulk_2h_mos2_hexagonal	6	1, 2	177.7 s	PASS
test_015_monoclinic_zro2	12	1, 3	719.7 s	PASS
test_016_exact_minimum_image_triclinic	n/a	none	0.7 s	PASS
test_017_nonorthogonal_polarisation_extxyz	2	1	10.9 s	PASS
test_018_nonorthogonal_vdw_graphene	2	1	7.1 s	PASS
test_019_nonorthogonal_npt_md	2	1	22.2 s	PASS
test_020_extreme_skew_sfc_mpi	n/a	1	40.3 s	PASS
test_021_hfo2_full_cell_npt	12	3	50.0 s	PASS
test_022_graphene_xy_npt	18	3	41.1 s	PASS
test_023_hcp_zr_method3_relax	2	1	192.5 s	PASS
test_024_rhombohedral_graphite_sfc_mpi	n/a	1, 2	30.2 s	PASS
test_025_blip_basis_smoke	8	2	2.5 s	PASS
test_026_nonorthogonal_blip_basis	8	1, 2	37.6 s	PASS
test_027_si_equivalent_cells	2	1	123.1 s	PASS
test_028_auto_full_stress	2	1	3.8 s	PASS
test_029_general_cell_sampling	2	1	3.4 s	PASS
test_030_triclinic_block_distance	n/a	none	0.5 s	PASS
test_031_rotated_triclinic_pdb	2	1	4.0 s	PASS
test_032_triclinic_surface_geometry	n/a	none	0.4 s	PASS

Test-by-test detail

test_001_bulk_Si_1proc_Diag

Status	PASS	Duration	2.60 s
Atoms	8	MPI ranks	2
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.08s
- PulayMixSC: Reached SCF tolerance of 0.91858E-06 after 5 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_001_bulk_Si_1proc_Diag.log

test_002_bulk_Si_1proc_OrderN

Status	PASS	Duration	9.93 s
Atoms	8	MPI ranks	2
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.04s

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_002_bulk_Si_1proc_OrderN.log

test_003_bulk_BTO_polarisation

Status	PASS	Duration	18.09 s
Atoms	5	MPI ranks	1
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.06s
- PulayMixSC: Reached SCF tolerance of 0.76963E-07 after 29 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_003_bulk_BTO_polarisation.log

test_004_isol_C2H4_4proc_PBE0CRI

Status	PASS	Duration	12.24 s
Atoms	6	MPI ranks	2
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.06s
- PulayMixSC: Reached SCF tolerance of 0.79157E-07 after 15 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_004_isol_C2H4_4proc_PBE0CRI.log

test_005_isol_C2H4_4proc_PBE0GTO

Status	PASS	Duration	56.43 s
Atoms	6	MPI ranks	2
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.06s
- PulayMixSC: Reached SCF tolerance of 0.78125E-07 after 14 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_005_isol_C2H4_4proc_PBE0GTO.log

test_006_isol_C2H4_4proc_PBE0ERI

Status	PASS	Duration	16.44 s
Atoms	6	MPI ranks	2
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.04s
- PulayMixSC: Reached SCF tolerance of 0.79157E-07 after 15 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_006_isol_C2H4_4proc_PBE0ERI.log

test_007_isol_CH_spinpol_1proc_PBE0CRI

Status	PASS	Duration	228.72 s
Atoms	2	MPI ranks	1
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.06s
- PulayMixSC: Reached SCF tolerance of 0.57025E-03 after 4 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_007_isol_CH_spinpol_1proc_PBE0CRI.log

test_008_surface_dipole

Status	PASS	Duration	4.49 s
Atoms	3	MPI ranks	1
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.04s
- PulayMixSC: Reached SCF tolerance of 0.70722E-08 after 38 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_008_surface_dipole.log

test_009_DFT_plus_U

Status	PASS	Duration	188.39 s
Atoms	2	MPI ranks	1
Entrypoint	direct CONQUEST run + pytest reference comparison	Exit code	0

Validation highlights

- 4 passed, 32 deselected in 0.04s
- SCF tolerance: 0.00000100
- PulayMixSC: Reached SCF tolerance of 0.85677E-06 after 29 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_009_DFT_plus_U.log

test_010_graphite_monoclinic

Status	PASS	Duration	194.05 s
Atoms	4	MPI ranks	1, 2
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- Workflow completed successfully!
- Error in process 1

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_010_graphite_monoclinic.log

test_011_hfo2

Status	PASS	Duration	63.23 s
Atoms	12	MPI ranks	1, 2
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- [1/4] Converged PBE/DZP SCF density on monoclinic HfO2
- SCF tolerance: 0.00001000
- new_SC_potl: Starting. Tolerance: 0.10000E-04
- PulayMixSC: Reached SCF tolerance of 0.99530E-05 after 7 iterations

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-235347/logs/test_011_hfo2.log

test_012_bulk_Si_primitive_nonorthogonal

Status	PASS	Duration	47.38 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- [2/5] Converged self-consistent density
- "initial_basis_error_bohr": 0.05996283515645341,
- "final_basis_error_bohr": 1.6081599726137713e-05

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_012_bulk_Si_primitive_nonorthogonal.log

test_013_rutile_TiO2_tetragonal

Status	PASS	Duration	153.66 s
Atoms	6	MPI ranks	1, 2
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights
- [2/5] Converged SCF density
Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_013_rutile_TiO2_tetragonal.log

test_014_bulk_2h_mos2_hexagonal

Status	PASS	Duration	177.73 s
Atoms	6	MPI ranks	1, 2
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- PASS: sulfur internal relaxation converged; MaxF 0.02736401 -> 0.00019562 Ha/a0.
- [2/5] Converged relaxed-cell SCF density

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_014_bulk_2h_mos2_hexagonal.log

test_015_monoclinic_zro2

Status	PASS	Duration	719.72 s
Atoms	12	MPI ranks	1, 3
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- [2/5] Symmetry restoration and converged SCF density

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_015_monoclinic_zro2.log

test_016_exact_minimum_image_triclinic

Status	PASS	Duration	0.73 s
Atoms	not applicable	MPI ranks	none
Entrypoint	run_exact_mic.sh	Exit code	0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_016_exact_minimum_image_triclinic.log

test_017_nonorthogonal_polarisation_extxyz

Status	PASS	Duration	10.92 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "extxyz_stress_error_gpa": 7.669131392162853e-07
- "maximum_quantum_error_e_per_bohr2": 2.053218706166149e-14,
- "maximum_extxyz_lattice_error_angstrom": 4.955079013768682e-09,
- "maximum_extxyz_stress_error_gpa": 7.927003740082483e-07
- "maximum_quantum_error_e_per_bohr2": 5e-12,
- "maximum_extxyz_lattice_error_angstrom": 1e-07,
- "maximum_extxyz_stress_error_gpa": 2e-05
- "failed_metrics": []

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_017_nonorthogonal_polarisation_extxyz.log

test_018_nonorthogonal_vdw_graphene

Status	PASS	Duration	7.14 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "energy_tolerance_ha": 0.002,
- "correction_tolerance_ha": 0.002

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_018_northogonal_vdw_graphene.log

test_019_nonorthogonal_npt_md

Status	PASS	Duration	22.17 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "md_frame_initial_error_bohr": 0.0,
- "initial_xsf_lattice_error_angstrom": 2.477539506884341e-09,
- "final_xsf_lattice_error_angstrom": 0.0,

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-235347/logs/test_019_northogonal_npt_md.log

test_020_extreme_skew_sfc_mpi

Status	PASS	Duration	40.30 s
Atoms	not applicable	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "sfc_lattice_normal_span_error_bohr": 2.1119547710668485e-07,
- "sfc_lattice_normal_span_error_bohr": 2.111954788830417e-07,
- "rank_energy_error_ha": 0.0,
- "rank_ewald_error_ha": 0.0,
- "basis_ewald_error_ha": 0.0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_020_extreme_skew_sfc_mpi.log

test_021_hfo2_full_cell_npt

Status	PASS	Duration	49.98 s
Atoms	12	MPI ranks	3
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "stress_symmetry_error": 0.0,

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_021_hfo2_full_cell_npt.log

test_022_graphene_xy_npt

Status	PASS	Duration	41.09 s
Atoms	18	MPI ranks	3
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "stress_symmetry_error": 1.000000000000001e-08,

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_022_graphene_xy_npt.log

test_023_hcp_zr_method3_relax

Status	PASS	Duration	192.46 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_023_hcp_zr_method3_relax.log

test_024_rhombohedral_graphite_sfc_mpi

Status	PASS	Duration	30.18 s
Atoms	not applicable	MPI ranks	1, 2
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- "sfc_lattice_normal_span_error_bohr": 2.7627839349619876e-07,
- "sfc_lattice_normal_span_error_bohr": 1.8575665716724643e-07,
- "rank_energy_error_ha": 3.481659405224491e-13,
- "rank_ewald_error_ha": 0.0,
- "rank_stress_error_gpa": 0.0,
- "representation_ewald_per_atom_error_ha": 3.666666970048027e-11

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_024_rhombohedral_graphite_sfc_mpi.log

test_025_blip_basis_smoke

Status	PASS	Duration	2.54 s
Atoms	8	MPI ranks	2
Entrypoint	run_workflow.sh	Exit code	0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_025_blip_basis_smoke.log

test_026_nonorthogonal_blip_basis

Status	PASS	Duration	37.61 s
Atoms	8	MPI ranks	1, 2
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights
- PASS: nonorthogonal blip value/derivative/inverse regression; energy=-33.692824094258 Ha; rank error=0.000e+00 Ha
Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_026_nonorthogonal_blip_basis.log

test_027_si_equivalent_cells

Status	PASS	Duration	123.09 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights
- PASS: equivalent diamond-Si cells; energy error=5.746e-06 Ha/2 atoms (0.0782 meV/atom); stress error=2.736e-02 GPa
Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-235347/logs/test_027_si_equivalent_cells.log

test_028_auto_full_stress

Status	PASS	Duration	3.79 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_028_au
to_full_stress.log

test_029_general_cell_sampling

Status	PASS	Duration	3.38 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- PASS: FFT cutoff sphere is contained and automatic k-point spacing uses reciprocal-vector norms

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_029_general_cell_sampling.log

test_030_triclinic_block_distance

Status	PASS	Duration	0.53 s
Atoms	not applicable	MPI ranks	none
Entrypoint	run_block_distance.sh	Exit code	0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_030_triclinic_block_distance.log

test_031_rotated_triclinic_pdb

Status	PASS	Duration	4.03 s
Atoms	2	MPI ranks	1
Entrypoint	run_workflow.sh	Exit code	0

Validation highlights

- maximum lattice error (Angstrom): 8.918e-06
- maximum fractional-coordinate error: 1.540e-04

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_031_rotated_triclinic_pdb.log

test_032_triclinic_surface_geometry

Status	PASS	Duration	0.42 s
Atoms	not applicable	MPI ranks	none
Entrypoint	run_surface_geometry.sh	Exit code	0

Log: /Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release/testsuite/test_runs/20260809-230103/logs/test_032_triclinic_surface_geometry.log

Provenance and method

Repository	/Users/augustin/Documents/CONQUEST-monoclinic-codex/CONQUEST-release
Branch	feature/general-nonorthogonal-cells
Commit	9248eb4de5f5
Upstream Test 005 fix	a8fceb6 - Initialize eri_gto to zero
Build	PASS (0.0 s)
Host	Lu-Laptop.local
Python	3.13.9

A PASS means the numbered test's canonical workflow returned zero and, for tests 001-009, the matching pytest reference comparisons also passed against a freshly written Conquest_out. The MPI guard re-read each active Conquest_input and coordinate file at launch time, so generated multi-stage workflows were constrained using their actual atom counts wherever available.